

Study Guide: Chapter 4 - Mechanical Oscillations

This guide covers the essential concepts, definitions, and formulas for the study of mechanical oscillations.

Part 1: Core Concepts & Vocabulary

First, you must understand the basic language of oscillations.

- Periodic Motion:** Any motion that repeats itself identically in a constant time interval.
- Oscillator:** A system that shows periodic motion.
- Equilibrium Position:** The stable position of the oscillator where the net force is zero (this is $x = 0$ for our spring).
- Elongation (x):** The position of the oscillator measured from the equilibrium position.
- Amplitude (X_m):** The maximum elongation (maximum displacement) from the equilibrium position. It is always a positive value.
- Period (T):** The time (in seconds, s) it takes to complete **one** full cycle.
 - $T = \frac{\text{total time}}{\text{number of cycles}} = \frac{\Delta t}{n}$
- Frequency (f):** The number of cycles completed in **one** second.
 - $f = \frac{\text{number of cycles}}{\text{total time}} = \frac{n}{\Delta t}$
- Period-Frequency Relation:** They are the inverse of each other.
 - $T = \frac{1}{f}$
 - $f = \frac{1}{T}$

Part 2: The 3 Modes & 3 Regimes of Oscillation

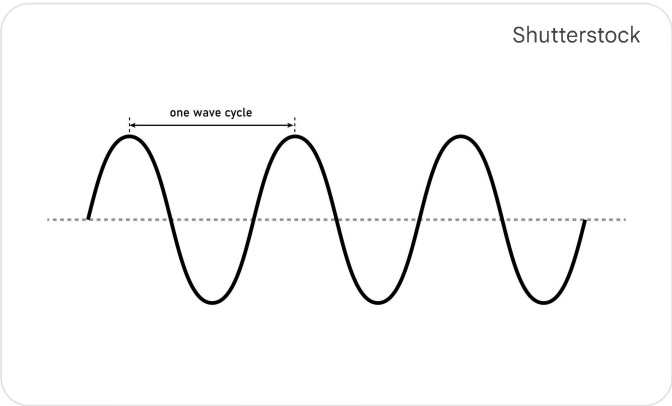
This section explains *how* an oscillator can behave based on friction and external forces.

The 3 Modes of Oscillation

- Free Oscillations:**
 - How it works:** You give the system energy *once* (e.g., pull a spring) and let it go.
 - Forces:** Only the restoring force (spring) and friction are acting.
- Maintained Oscillations:**
 - How it works:** A free oscillation where an external mechanism *replaces* the energy lost to friction in each cycle.
 - Goal:** To keep the amplitude constant, just like in an ideal (no friction) system.
 - Example:** A pendulum clock, where a battery or weights provide energy to counteract air resistance.
- Forced Oscillations:**
 - How it works:** An external, periodic force is *continuously* applied to the oscillator, forcing it to oscillate at the driver's frequency.

The 3 Regimes of *Free* Oscillation (Based on Friction)

This is the most important conceptual part. The regime is determined by the *amount* of friction.

Regime	Amount of Friction	Amplitude (X_m)	Mechanical Energy (ME)	Graph Shape
Periodic	None (Ideal)	Constant	Conserved ($ME = \text{const}$)	Shutterstock 
Pseudo-Periodic	Low (Realistic)	Decreases over time	Decreases (Lost to friction)	A "dying" sine wave.

Aperiodic	Very High	No oscillation	Decreases (Lost to friction)	No wave, just a slow return to equilibrium.
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Part 3: Analysis of Simple Harmonic Motion (SHM)

This is the detailed mathematical study of the **Periodic Regime** (no friction).

1. The Differential Equation (The "Rule" of the Motion)

This equation defines SHM. We can get it in two ways:

- Dynamic Method (Newton's 2nd Law):**
 - $\Sigma \vec{F} = m\vec{a}$
 - The only horizontal force is the spring's restoring force: $\vec{T}$
 - By Hooke's Law: $T = -Kx$
 - So, $-Kx = ma \implies -Kx = mx''$
 - Rearranging gives the equation: $x'' + \frac{K}{m}x = 0$
- Energetic Method (Energy Conservation):**
 - In this ideal regime, $ME = \text{constant}$.
 - $ME = KE + EPE = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2$
 - Because ME is constant, its derivative with respect to time is zero: $\frac{d(ME)}{dt} = 0$
 - $\frac{d}{dt} \left(\frac{1}{2}mv^2 + \frac{1}{2}Kx^2 \right) = 0$
 - Using the chain rule: $mva + Kxv = 0 \implies v(ma + Kx) = 0$
 - Since v is not always zero, we get: $ma + Kx = 0 \implies mx'' + Kx = 0$
 - This gives the same equation: $x'' + \frac{K}{m}x = 0$

2. Proper Pulsation (ω_0)

This is a new key term that simplifies the differential equation.

- Definition:** We define $\omega_0^2 = \frac{K}{m}$.
- Formula:** $\omega_0 = \sqrt{\frac{K}{m}}$
- The Equation (Short Form):** $x'' + \omega_0^2x = 0$

3. The Solution (The Equation of x vs. t)

The solution to the differential equation tells us the position x at any time t .

$$x(t) = X_m \sin(\omega_0 t + \varphi)$$

- $x(t)$: Elongation (position) at time t .
- X_m : Amplitude (determined by initial energy or displacement).
- ω_0 : Proper Pulsation (in rad/s). This is $\sqrt{K/m}$.
- $(\omega_0 t + \varphi)$: The "phase" at time t (in rad).
- φ : The "initial phase" (in rad). This value tells the oscillator where it was at $t = 0$. You must solve for it using initial conditions (x_0 and v_0).

4. Velocity and Acceleration

We get $v(t)$ and $a(t)$ by taking derivatives of $x(t)$.

- Velocity $v(t)$:**
 - $v(t) = x'(t) = \frac{d}{dt} [X_m \sin(\omega_0 t + \varphi)]$
 - $v(t) = X_m \omega_0 \cos(\omega_0 t + \varphi)$
 - Maximum Speed V_{max} : $V_{max} = X_m \omega_0$ (This happens when $x = 0$, at equilibrium)
- Acceleration $a(t)$:**
 - $a(t) = v'(t) = \frac{d}{dt} [X_m \omega_0 \cos(\omega_0 t + \varphi)]$
 - $a(t) = -X_m \omega_0^2 \sin(\omega_0 t + \varphi)$
 - Notice that $X_m \sin(\omega_0 t + \varphi)$ is just $x(t)$.
 - This gives the relationship: $a(t) = -\omega_0^2 x(t)$ (This is the *definition* of SHM in many textbooks!)

Part 4: Energy in SHM (Undamped)

- Elastic Potential Energy (EPE): Stored in the spring.
 - $EPE = \frac{1}{2}Kx^2$
 - $EPE(t) = \frac{1}{2}K [X_m \sin(\omega_0 t + \varphi)]^2 = \frac{1}{2}K X_m^2 \sin^2(\omega_0 t + \varphi)$
- Kinetic Energy (KE): Energy of motion.
 - $KE = \frac{1}{2}mv^2$
 - $KE(t) = \frac{1}{2}m [X_m \omega_0 \cos(\omega_0 t + \varphi)]^2 = \frac{1}{2}m \omega_0^2 X_m^2 \cos^2(\omega_0 t + \varphi)$
 - Since $K = m\omega_0^2$: $KE(t) = \frac{1}{2}K X_m^2 \cos^2(\omega_0 t + \varphi)$
- Mechanical Energy (ME): Total energy. It is **CONSERVED**.
 - $ME = KE + EPE$
 - $ME = \frac{1}{2}K X_m^2 \cos^2(\omega_0 t + \varphi) + \frac{1}{2}K X_m^2 \sin^2(\omega_0 t + \varphi)$
 - $ME = \frac{1}{2}K X_m^2 [\cos^2(\dots) + \sin^2(\dots)]$
 - Since $\cos^2(\theta) + \sin^2(\theta) = 1$, the total energy is **constant**: $ME = \frac{1}{2}K X_m^2$

Energy Trade-off:

- At the center ($x = 0$):
 - $EPE = 0$ (no stretch)
 - $KE = KE_{max} = ME$
 - Speed is V_{max}
- At the extremes ($x = \pm X_m$):
 - $EPE = EPE_{max} = \frac{1}{2}K X_m^2 = ME$
 - $KE = 0$ (stops for an instant)
 - Speed is $v = 0$

Part 5: Formula Summary Sheet

Core Definitions:

- $T = 1/f$
- $f = 1/T$

Properties of the Oscillator (SHM):

- $\omega_0 = \sqrt{\frac{K}{m}}$
- $T_0 = \frac{2\pi}{\omega_0} = 2\pi \sqrt{\frac{m}{K}}$
- $f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{K}{m}}$

Equations of Motion (SHM):

- $x(t) = X_m \sin(\omega_0 t + \varphi)$
- $v(t) = X_m \omega_0 \cos(\omega_0 t + \varphi)$
- $a(t) = -X_m \omega_0^2 \sin(\omega_0 t + \varphi) = -\omega_0^2 x(t)$

Max Values (SHM):

- $V_{max} = X_m \omega_0$
- $a_{max} = X_m \omega_0^2$ (occurs at $x = \pm X_m$)

Energy (SHM):

- $EPE = \frac{1}{2}Kx^2$
- $KE = \frac{1}{2}mv^2$
- $ME_{total} = \frac{1}{2}K X_m^2 = \frac{1}{2}m V_{max}^2 = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2 = \text{Constant}$